EDA_Titanic

July 6, 2026

1 Task 5: Exploratory Data Analysis (EDA)

1.1 Elevate Labs Data Analyst Internship

1.1.1 Objective

The objective of this task is to perform Exploratory Data Analysis (EDA) on the Titanic dataset using Python libraries such as Pandas, Matplotlib, and Seaborn to discover patterns, relationships, and insights.

1.1.2 Tools Used

- Python
- Pandas
- Matplotlib
- Seaborn
- NumPy

---

[1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

%matplotlib inline
print("Libraries installed successfully")
```

```
Libraries installed successfully
```

[4]:
```
df = pd.read_csv("train.csv")

df.head()
```

[4]:
```
   PassengerId  Survived  Pclass  \
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3
```

```
                                                Name     Sex   Age  SibSp  \
0                            Braund, Mr. Owen Harris    male  22.0      1
1  Cumings, Mrs. John Bradley (Florence Briggs Th…  female  38.0      1
2                             Heikkinen, Miss. Laina  female  26.0      0
3       Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1
4                           Allen, Mr. William Henry    male  35.0      0

   Parch            Ticket     Fare Cabin Embarked
0      0         A/5 21171   7.2500   NaN        S
1      0          PC 17599  71.2833   C85        C
2      0  STON/O2. 3101282   7.9250   NaN        S
3      0            113803  53.1000  C123        S
4      0            373450   8.0500   NaN        S
```

[6]:
```python
print("Rows : " , df.shape[0])
print("Columns : "  , df.shape[1])
```

```
Rows :  891
Columns :  12
```

#The dataset contains 891 rows and 12 columns.

[11]:
```python
df.columns
```

[11]:
```
Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp',
       'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked'],
      dtype='str')
```

[12]:
```python
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    str
 4   Sex          891 non-null    str
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    str
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    str
 11  Embarked     889 non-null    str
dtypes: float64(2), int64(5), str(5)
```

```
memory usage: 83.7 KB
```

#The dataset contains numerical and categorical features. #Some columns contain missing values.

[13]:
```python
df.describe()
```

[13]:
```
       PassengerId    Survived      Pclass         Age       SibSp  \
count   891.000000  891.000000  891.000000  714.000000  891.000000
mean    446.000000    0.383838    2.308642   29.699118    0.523008
std     257.353842    0.486592    0.836071   14.526497    1.102743
min       1.000000    0.000000    1.000000    0.420000    0.000000
25%     223.500000    0.000000    2.000000   20.125000    0.000000
50%     446.000000    0.000000    3.000000   28.000000    0.000000
75%     668.500000    1.000000    3.000000   38.000000    1.000000
max     891.000000    1.000000    3.000000   80.000000    8.000000

            Parch        Fare
count  891.000000  891.000000
mean     0.381594   32.204208
std      0.806057   49.693429
min      0.000000    0.000000
25%      0.000000    7.910400
50%      0.000000   14.454200
75%      0.000000   31.000000
max      6.000000  512.329200
```

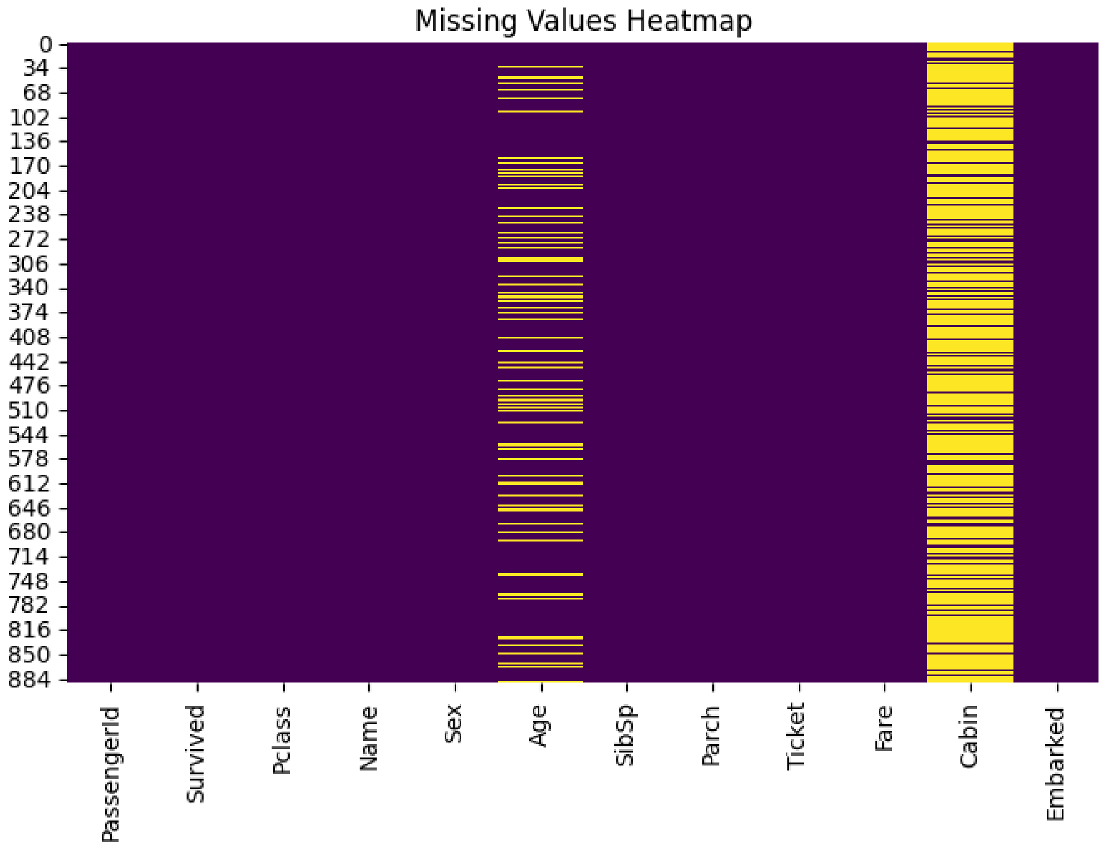
#The statistical summary provides information such as mean, #standard deviation, minimum, maximum and quartiles for numerical columns.

[8]:
```python
df.duplicated().sum()
```

[8]:
```
np.int64(0)
```

#there are no duplicates in the dataset

[14]:
```python
plt.figure(figsize=(8,5))

sns.heatmap(df.isnull(),
            cbar=False,
            cmap="viridis")

plt.title("Missing Values Heatmap")
plt.show()
```



#Age, Cabin and Embarked contain missing values. #Cabin has the highest number of missing records.

[9]:
```
df.isnull().sum()
```

[9]:
```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

[15]:
```
df['Age'] = df['Age'].fillna(df['Age'].median())

df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])

df['Cabin'] = df['Cabin'].fillna("Unknown")
```

[16]:
```
df.isnull().sum()
```

[16]:
```
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Ticket         0
Fare           0
Cabin          0
Embarked       0
dtype: int64
```

[37]:
```
#All missing values have been handled successfully.
```

[17]:
```
df.dtypes
```

[17]:
```
PassengerId      int64
Survived         int64
Pclass           int64
Name               str
Sex                str
Age            float64
SibSp            int64
Parch            int64
Ticket             str
Fare           float64
Cabin              str
Embarked           str
dtype: object
```

[18]:
```
df['Survived'].value_counts()
```

[18]:
```
Survived
0    549
1    342
Name: count, dtype: int64
```

[25]:
```python
survival_rate = df['Survived'].value_counts(normalize=True)*100
print(survival_rate)
```

```
Survived
0    61.616162
1    38.383838
Name: proportion, dtype: float64
```

#The percentage of passengers who survived is lower than those who did not survive.

[26]:
```python
df.to_csv("Titanic_Cleaned.csv", index=False)
```

1.2 Data Cleaning Summary

Imported Dataset

Checked Dataset Shape

Examined Data Types

Generated Statistical Summary

Identified Missing Values

Removed Duplicate Records

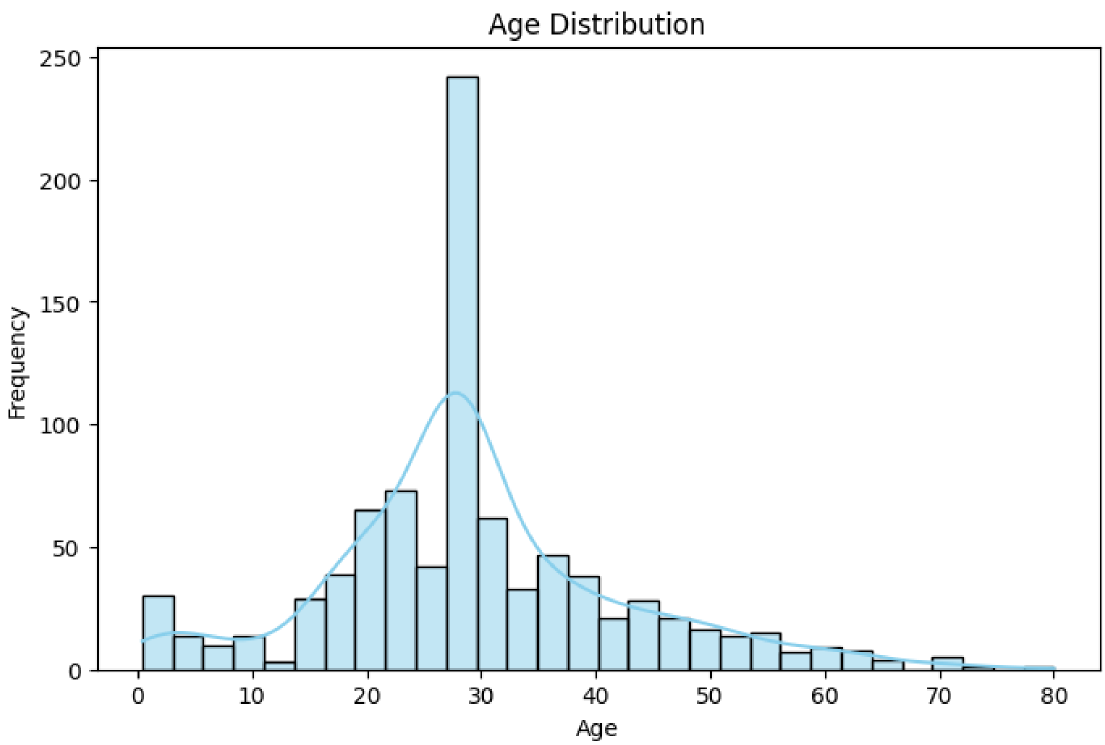
Filled Missing Values

Exported Clean Dataset

The dataset is now ready for Exploratory Data Analysis.

1.3 Age Distribution

[35]:
```python
plt.figure(figsize=(8,5))

sns.histplot(df['Age'], bins=30, kde=True, color='skyblue')

plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Frequency")

plt.show()
```

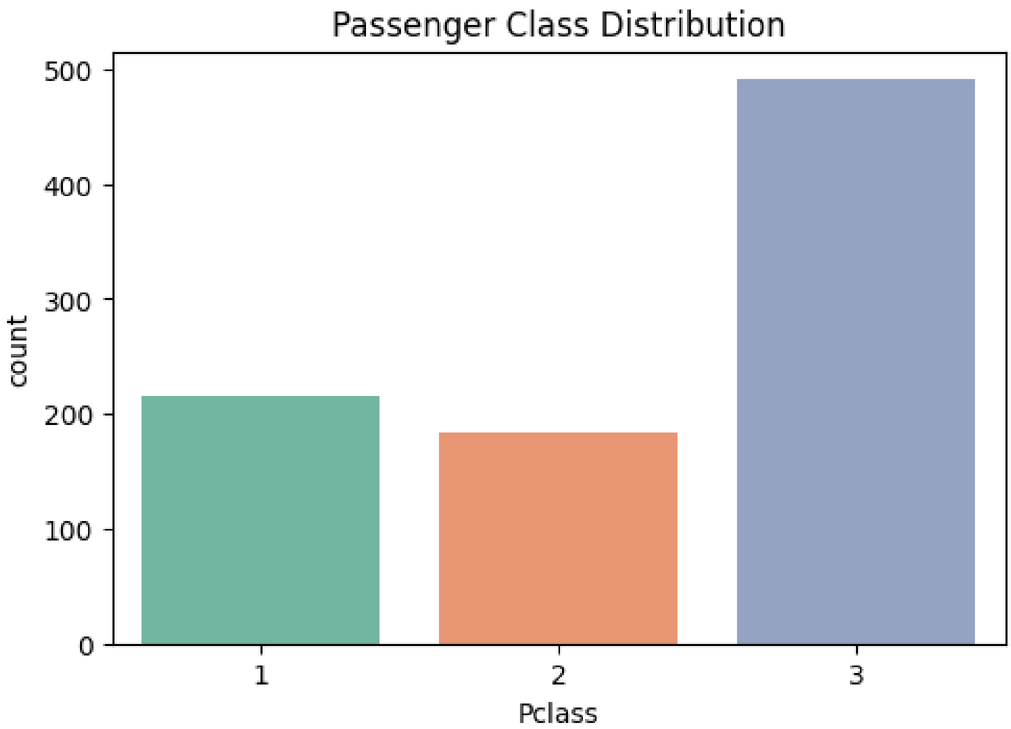


#Most passengers were between 20 and 40 years of age. #The age distribution is slightly right-skewed with fewer elderly passengers.

1.3.1 Passenger Class Distribution

[44]:
```
plt.figure(figsize=(6,4))

sns.countplot(data=df, x='Pclass', palette='Set2')

plt.title("Passenger Class Distribution")

plt.show()
```

```
C:\Users\sohel\AppData\Local\Temp\ipykernel_2756\1042294632.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in
v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same
effect.

  sns.countplot(data=df, x='Pclass', palette='Set2')
```

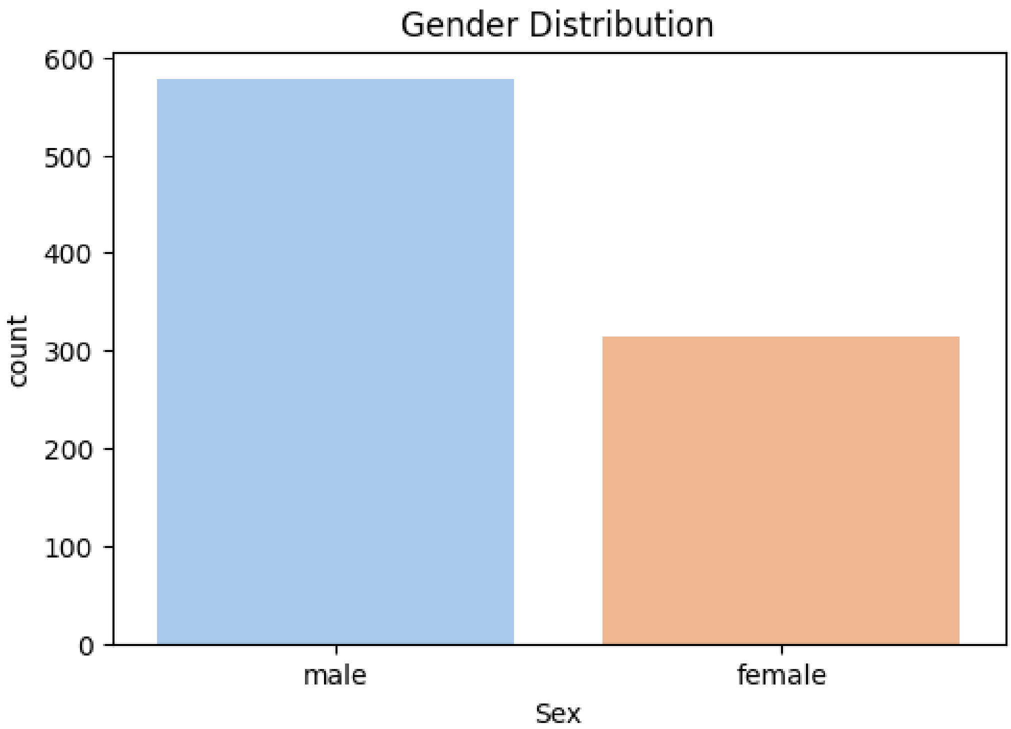


Most passengers belonged to the Third Class, followed by First Class and Second Class.

1.3.2 Gender Distribution

[46]:
```
plt.figure(figsize=(6,4))

sns.countplot(data=df, x='Sex', palette='pastel')

plt.title("Gender Distribution")

plt.show()
```

```
C:\Users\sohel\AppData\Local\Temp\ipykernel_2756\260862961.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in
v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same
effect.

  sns.countplot(data=df, x='Sex', palette='pastel')
```

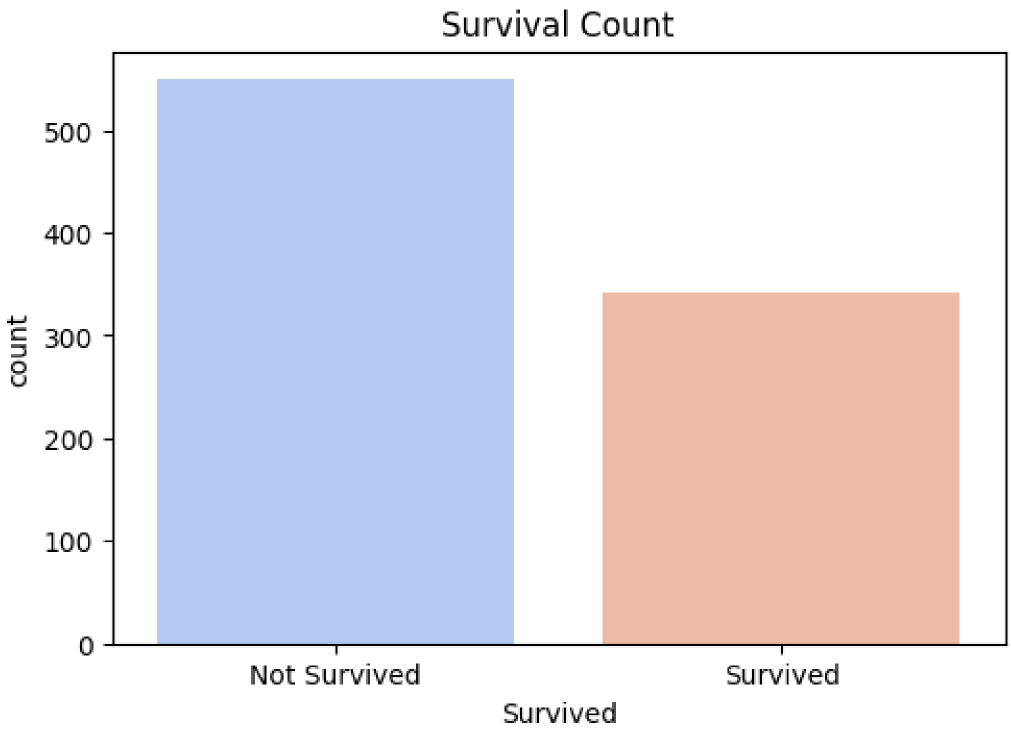


#The dataset contains more male passengers than female passengers.

1.3.3 Survival Count

[47]:
```
plt.figure(figsize=(6,4))

sns.countplot(data=df, x='Survived', palette='coolwarm')

plt.title("Survival Count")

plt.xticks([0,1],["Not Survived","Survived"])

plt.show()
```

```
C:\Users\sohel\AppData\Local\Temp\ipykernel_2756\2155088975.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in
v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same
effect.

  sns.countplot(data=df, x='Survived', palette='coolwarm')
```

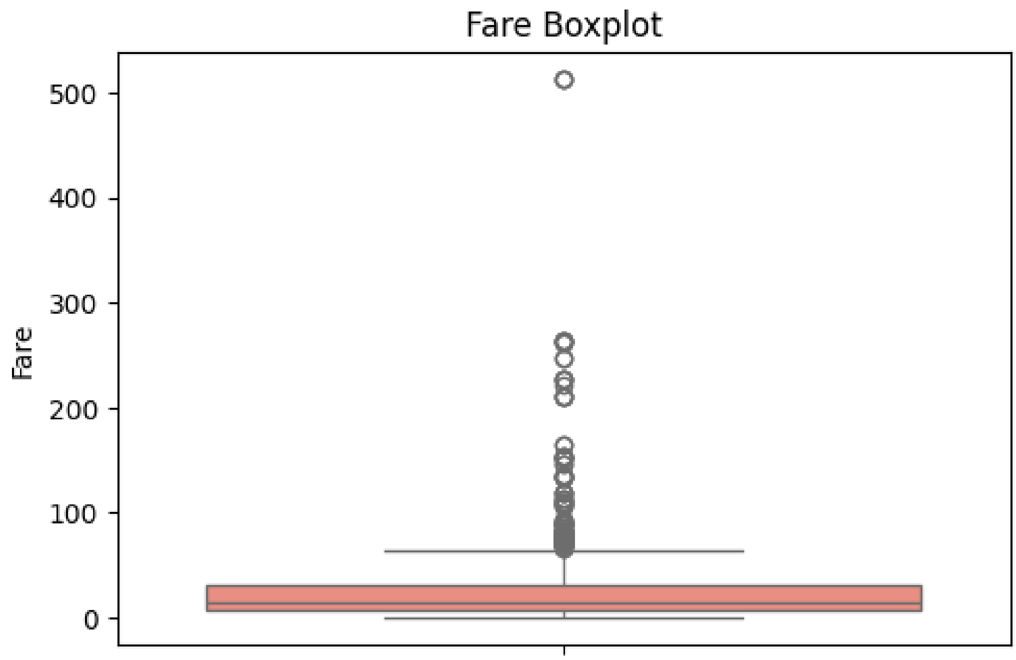


2 The number of passengers who did not survive

3 is greater than the number of survivors.

3.0.1 Boxplot of Fare

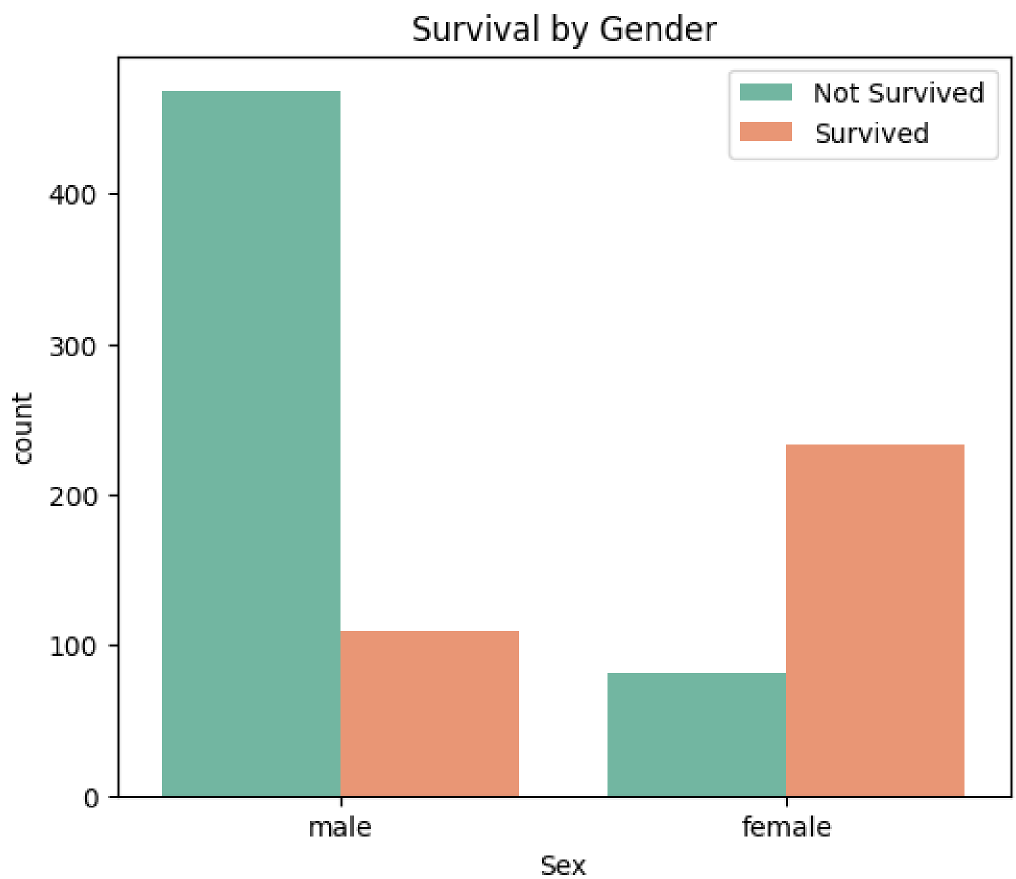
[48]:
```
plt.figure(figsize=(6,4))

sns.boxplot(y=df['Fare'], color='salmon')

plt.title("Fare Boxplot")

plt.show()
```



#Several high-fare outliers are present, #indicating a few passengers paid significantly higher ticket prices.

3.0.2 Survival by Gender

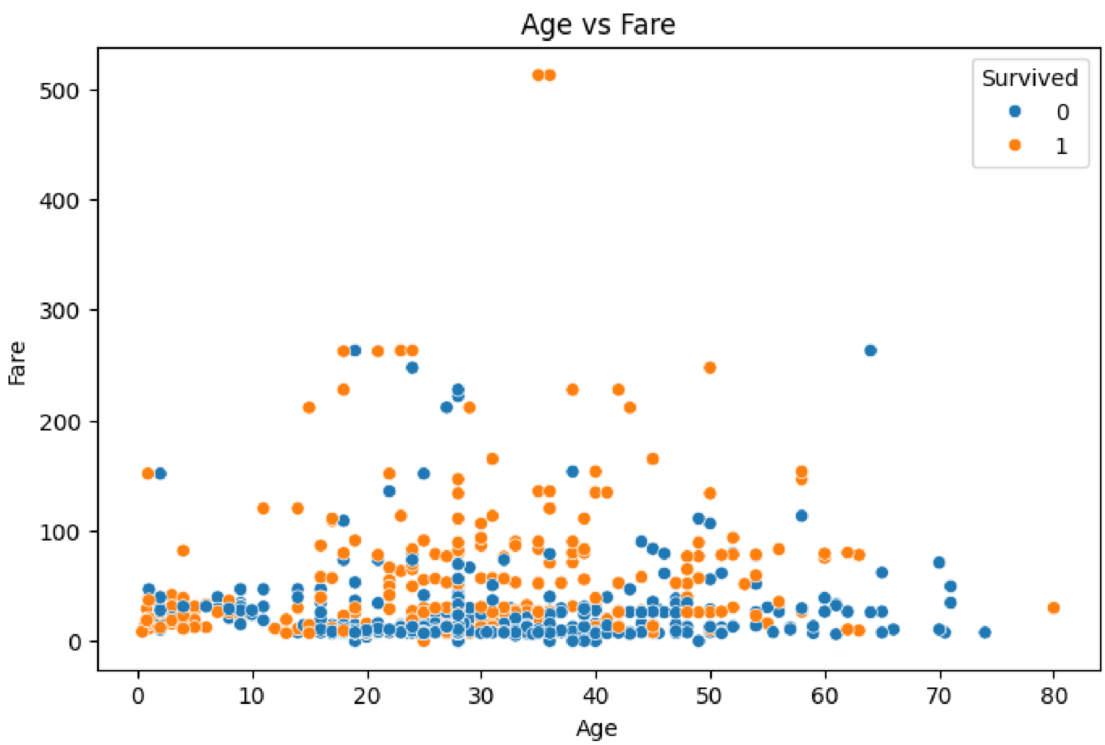
[49]:
```
plt.figure(figsize=(6,5))

sns.countplot(data=df,
              x='Sex',
              hue='Survived',
              palette='Set2')

plt.title("Survival by Gender")

plt.legend(['Not Survived','Survived'])

plt.show()
```



#Female passengers had a significantly higher survival rate than male passengers, indicating that gender played an important role in survival.

3.0.3 Scatter plot Age vs Fare

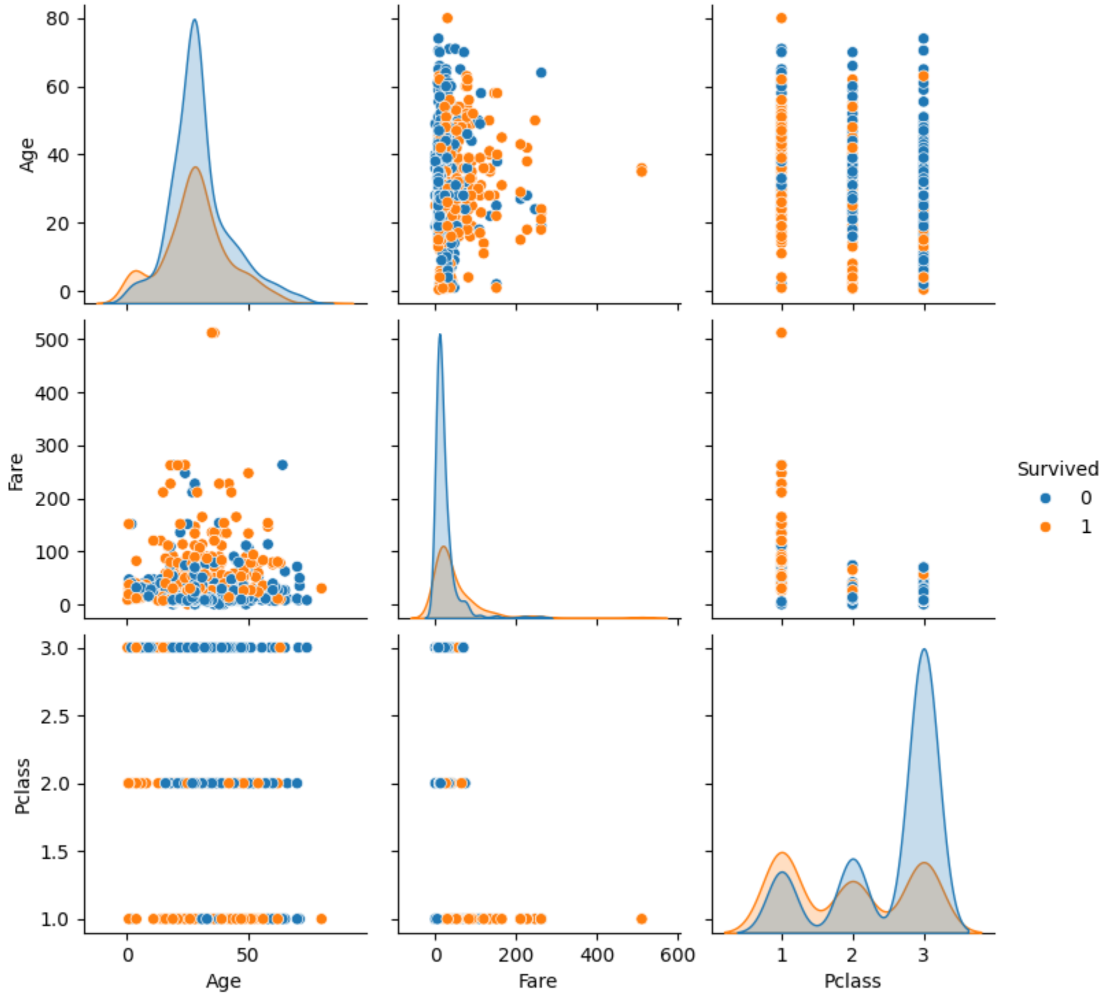
[50]:
```
plt.figure(figsize=(8,5))

sns.scatterplot(data=df,
                x='Age',
                y='Fare',
                hue='Survived')

plt.title("Age vs Fare")

plt.show()
```



#Passengers who paid higher fares were more likely to survive. No strong relationship exists between age and fare.

3.0.4 Pairplot

[51]:
```
sns.pairplot(df[['Age',
                 'Fare',
                 'Pclass',
                 'Survived']],

             hue='Survived')

plt.show()
```

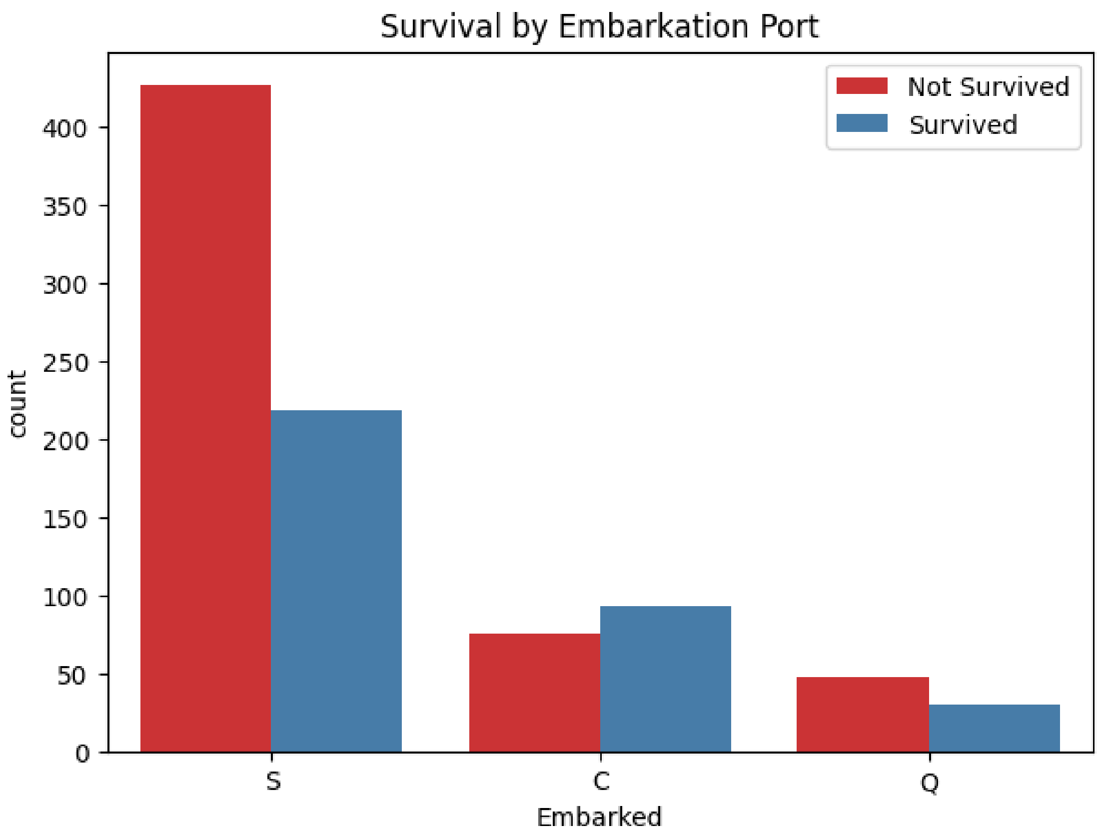


#The pairplot provides an overview of relationships among numerical variables. Survival patterns become more visible when comparing fare and passenger class.

3.0.5 Survival by Embarkation Port

[53]:
```
plt.figure(figsize=(7,5))

sns.countplot(data=df,
              x='Embarked',
              hue='Survived',
              palette='Set1')

plt.title("Survival by Embarkation Port")

plt.legend(['Not Survived','Survived'])
```

```
plt.show()
```



3.1 Bivariate & Multivariate Analysis Summary

The following relationships were identified:

- Female passengers had a much higher survival rate.
- First-class passengers were more likely to survive.
- Higher ticket fares were associated with higher survival.
- Passenger class strongly influenced ticket prices.
- Most numerical variables showed weak correlations.
- Passenger class and fare were the most influential variables in explaining survival.

These findings help understand the interactions between different variables and reveal the factors influencing passenger survival.

4 Key Findings

After performing Exploratory Data Analysis on the Titanic dataset, the following insights were obtained:

1. Most passengers belonged to the Third Passenger Class.

2. Male passengers outnumbered female passengers.
3. Female passengers had a significantly higher survival rate.
4. First-class passengers were more likely to survive.
5. Most passengers were between 20 and 40 years old.
6. Fare distribution was highly right-skewed.
7. Higher ticket fares were generally associated with higher survival rates.
8. Southampton was the most common embarkation port.
9. Passenger class strongly influenced fare and survival.
10. Most numerical variables showed weak correlations except for fare and passenger class.
11. Missing values were successfully handled before analysis.
12. The dataset was clean and suitable for visualization and statistical exploration.

5 Learning Outcomes

Through this task, I learned:

- Data Exploration using Pandas
- Data Cleaning Techniques
- Handling Missing Values
- Descriptive Statistics
- Univariate Analysis
- Bivariate Analysis
- Multivariate Analysis
- Correlation Analysis
- Data Visualization using Matplotlib
- Statistical Visualization using Seaborn
- Drawing meaningful business insights from data

6 Export Notebook

After completing the notebook:

1. Click **File**
2. Select **Save and Checkpoint**
3. Choose **Download As**
4. Select **PDF (.pdf)**

If PDF export is unavailable:

- File → Print
- Save as PDF

OR

Convert using:

jupyter nbconvert --to pdf Task5_EDA_Titanic.ipynb

```
[ ]:
```